

# From Genes to “Temes”: The Evolution of Memetics in Biological and Digital Realms

## Introduction

In 1976, evolutionary biologist Richard Dawkins introduced the term *meme* to describe a unit of cultural transmission or imitation. Just as genes are the replicators of biological evolution, Dawkins suggested that memes – ideas, tunes, phrases, fashions, and other bits of information – act as replicators that drive cultural evolution. This bold proposal laid the groundwork for *memetics*, the study of how memes propagate, by reframing culture in Darwinian terms. Over the decades, memetics has expanded to intersect with new domains: the rise of **digital memes** (Internet-borne memes) demonstrates how information can spread rapidly in the digital age, and a growing body of theoretical work explores potential links between biology, computation, and consciousness. This essay synthesizes these strands: it examines the evolution of memetics, distinguishes between biological and digital memes, and surveys the *Cognitive-Computational Substrate (CCS)* and *Nested Set System (NSS)* framework as a means of understanding memes within a self-organizing, informational reality.

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## The Evolution of Memetics: From Dawkins to Dennett

Dawkins’ introduction of memes extended evolutionary principles—variation, competition, and selection—beyond the genetic sphere, arguing that any replicator exhibiting heredity, mutation, and selection would undergo a Darwinian process. Cultural ideas fit this description: they persist by capturing human attention, mutate when passed through different minds, and face selection pressures determined by social acceptance and emotional resonance. Dawkins also likened memes to “parasites” infecting human minds, suggesting that once memes existed as replicators, they could follow an evolutionary trajectory independent of genetic evolution. This perspective ignited debate about the autonomy of cultural evolution, particularly when memes could, in principle, run counter to genetic interests (e.g., celibacy-promoting religious memes).

Philosopher Daniel Dennett later championed these ideas by emphasizing that evolution is a *substrate-neutral* algorithm. In his view, biology is merely one medium in which Darwinian processes operate. Culture, he argued, meets the same criteria for evolutionary change: memes, like genes, compete for proliferation in a finite resource space (our minds). Dennett provocatively proposed that human consciousness itself might be largely shaped by, or even composed of, memes, illustrating the extent to which these replicators can restructure the mental landscape.

Despite their compelling nature, memes were initially dismissed by some scholars as “mythical entities” or mere metaphors. Many cognitive scientists and anthropologists, for example, attempted to keep culture subordinate to genetic evolution (genes biasing cultural outcomes) rather than treat memes as autonomous replicators. It was not until the Internet era—with viral memes accelerating cultural

diffusion—that memetics began receiving broader attention, as digital connectivity showcased the replicative power of these cultural units in real time.

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## Biological Memes and Digital Memes: An Interaction

Although memes were originally defined broadly (e.g., a tune, idea, practice), today the term “meme” often evokes online images, hashtags, or videos that “go viral.” These **digital memes** indeed epitomize the memetic process—replicating, mutating, and competing for attention. The digital sphere amplifies meme dissemination to a speed and scale never before possible. Notably, Internet memes have influenced political discourse and social movements. Analysts have argued that memes contributed to phenomena like the rise of the alt-right movement and the 2016 U.S. presidential election, revealing how quickly memetic evolution can shape real-world outcomes.

Psychologist Susan Blackmore introduced the notion that technology has birthed a **third replicator**. Beyond genes and memes, “temes” (or “tremes”) are replicators that exist purely in digital or machine-driven environments. Earth thus hosts three layers of evolutionary dynamics: gene-based biological evolution, meme-based cultural evolution, and teme-based technological evolution. Blackmore observes that smartphones, social platforms, and AI-driven content generation have accelerated and diversified the replicative capabilities of these digital entities, raising the possibility that artificial memes (temes) could ultimately evolve beyond human oversight.

At present, biological and digital memes interact symbiotically. Human consciousness provides meaning and interpretation to digital memes, while online networks drastically increase the reach and speed of memetic spread. However, distinguishing between a meme as “information” and a meme as “meaning” is crucial. While a computer can copy an image or text string exactly—thereby replicating the meme’s data—it does not, according to philosopher Federico Faggin, *understand* that data in a subjective, experiential way. The richness of bio-memes (ideas in conscious minds) includes intentions and emotional resonance; by contrast, digital memes on servers lack first-person qualia. This highlights the importance of the substrate within which memes reside, be it biological brain or digital mechanism.

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## Cognitive-Computational Substrate (CCS): Bridging Mind and Machine

The concept of a *Cognitive-Computational Substrate* posits that reality itself might be undergirded by a fundamental information-processing layer, one that spans neural and digital systems alike. Several thinkers propose ideas aligning with such a substrate:

- **Christopher Langan’s CTMU** frames reality as a self-processing, self-referential “language” of information. In this view, “infocognition” is the basic stuff of existence, and memes are distinct configurations of that infocognitive medium.

- **Donald Hoffman's Conscious Realism** suggests that what we perceive (space, time, physical objects) is akin to a user interface. He envisages a web of “conscious agents” that underlies the apparent physical world. A meme, then, is a pattern of information that passes from agent to agent, each agent “rendering” it according to its interface.

In this spirit, a universal substrate may unify neurons (in biological minds) and transistors (in computers) as different physical implementations of underlying computation. Memes, defined as replicable informational units, can move across these implementations. A meme might start as a creative spark in a human mind, be encoded digitally, spread through global networks, and then re-enter other minds, continuing an endless cycle of replication.

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## Nested Set System (NSS): Hierarchies of Information

Alongside the CCS concept is the *Nested Set System* (NSS). Visualize a series of Russian nesting dolls—each smaller doll contained within a larger one, each reflecting the structure of the whole. Applied to memetics, this framework posits that reality contains hierarchies of informational domains (e.g., cells, brains, societies, global communication networks). Genes are nested in organisms, which carry memes in their minds, which in turn produce technology that hosts memes. Each level piggybacks on the prior one, forming a fractal-like system. By recognizing nested hierarchies:

1. **Genes** shape our brains.
2. **Memes** arise and propagate in these brains (human culture).
3. **Temes** leverage human tools and networks (digital culture).

Observing this nesting clarifies that memetic competition occurs not just in individuals, but also in large-scale social and technological contexts. Karl Friston's Free Energy Principle (FEP) further suggests that all cognitive systems strive to minimize surprise. Memes that align well with a society's existing beliefs or reduce collective uncertainty may spread more easily. Conversely, memes introducing high “surprise” risk rejection or suppression. This interplay of resonance and resistance can be studied at each nested layer—neural, social, and digital—highlighting the multi-level dance of memetic evolution.

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## Theoretical Convergence and Implications

Memetics stands at the crossroads of evolutionary biology, cognitive science, AI, and philosophy of mind. Darwinian models help explain the replication, mutation, and selection of cultural ideas. Neural and perceptual frameworks (e.g., Hoffman's user-interface model, Friston's Free Energy Principle) show how brains filter which memes stick or mutate. Philosophical arguments (e.g., Langan's CTMU, Faggin's emphasis on consciousness) press us to consider the fundamental reality in which these replicative processes unfold.

### 1. **Human Creativity and Agency**

Memetics prompts a question: Are we originators of ideas, or do memes merely use us as replication vehicles? Likely both—humans actively remix, reshape, and combine memes, but once birthed, these memes follow an evolutionary trajectory that can supersede their creators' intentions.

### 2. **Information Ecosystems**

Digital memes demonstrate the speed and scale at which information can spread, often exploiting cognitive biases. Societies could develop “memetic immune systems,” combining platform design, algorithmic checks, and personal literacy skills to filter harmful or manipulative memes. Conversely, beneficial or “prosocial” memes—like public health campaigns—could be carefully engineered for maximum constructive impact.

### 3. **Bridging Mind and Matter**

By viewing reality as a cognitive-computational substrate, memes become the *software* bridging *wetware* (brains) and *hardware* (machines). This angle reframes classic mind-body debates into questions of how informational patterns manifest differently in neuronal versus digital circuits, yet follow consistent laws of replication and adaptation.

### 4. **Future Evolution**

If machines ever develop genuine consciousness, they might generate memes beyond human imagining. Alternatively, even unconscious AI might replicate “super-memes” at speeds outpacing human capacity to adapt. The notion of a new replicator—*temes*—raises both hope for immense creative leaps and fear of losing human autonomy in a flood of machine-propagated information.

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## Conclusion

From Dawkins' early depiction of memes as self-serving cultural replicators to modern explorations of digitally supercharged “*temes*,” memetics has proved both controversial and remarkably prescient. Its relevance is heightened in an age when ideas can traverse the globe instantly, sometimes reshaping entire social landscapes. The hypothesis of a Cognitive-Computational Substrate suggests that the boundary between biology and technology, and even between mind and matter, may be more permeable than previously assumed; memes comfortably cross these domains, thriving wherever the conditions for replication exist.

Likewise, the Nested Set System reveals the hierarchical nature of information flow, illustrating that memes nest within brains, societies, and technological infrastructures. Understanding the interplay of bio-memes and digi-memes at different levels could help avert memetic overload, guide us in designing healthier information ecosystems, and even prepare us for scenarios in which AI-driven memes gain prominence.

Ultimately, memetics not only asks us to see ourselves as shaped by the ideas we carry but also to embrace our role as active participants in a grand evolutionary game of information. By studying how

memes evolve, the substrates they occupy, and the nested systems that shape them, we inch closer to comprehending our own collective mind—and possibly, the emergent mind of our machines.

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